Design for Dis/Ability: A Crip Inquiry into Personal Energy Tracking

Irene Kaklopoulou
Department of Informatics
Umeå University
Umeå, Sweden
eirini.kaklopoulou@umu.se

Sarah Homewood
Department of Computer Science
University of Copenhagen
Copenhagen, Denmark
sfh@di.ku.dk

Pedro Sanches
Department of Informatics
Umeå University
Umeå, Sweden
pedro.sanches@umu.se

Abstract

This study explores the phenomenon of energy and self-tracking technologies, moving beyond the context of managing chronic conditions. Our approach to designing from the experiences of people with disability is informed by crip theory, which challenges societal norms of health and ability. We analysed 50 survey responses and 15 interviews with wearable tracker users and found that self-tracking shapes interpretation of energy and self-care strategies. Our findings indicate that tracking significantly affect perceptions and judgments of bodily activity, energy and rest. We found a notable disconnect between the metrics provided by the trackers and the subjective understanding of personal energy meanings, especially during events of bodily and contextual changes such as travelling, illness, or menstrual cycle. This research contributes to discourses on energy in self-tracking technologies and advocates for designing more inclusive, crip futures for everyone that celebrate irregularity, fluctuation, and change, accommodating diverse bodily rhythms in energy tracking practices.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**.

Keywords

energy, crip theory, wellbeing, metrics, self-tracking, biodata, fitness tracking, fatigue, energy management, quantified self

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1 Introduction

Quantification methods through biodata have become routine aspects of self-knowledge for many [77, 107]. Over the last two decades, self-tracking technologies have transitioned from simply counting steps and estimating calories towards providing complex "derived" metrics that purportedly rate body energy based on how ready bodies are for fitness training, how much recovery they need, as well as classifying sleep stages and stress metrics [16]. Self-tracking practices have thereby emerged around these metrics, centred around managing and evaluating various spheres of everyday lives, including fitness, leisure, performance, or health more generally [5, 16, 80].

The aim of our paper is to study how self-tracking metrics contribute to individual conceptualisations of *energy levels* [116]. Tracking energy is a common practice within the disability and neurodivergent community, where fatigue and post-exertion malaise are common symptoms [47, 91]. Many individuals and communities with chronic conditions and/or disabilities adopt and repurpose commercially available wearable fitness tracking devices for objectives not intentionally designed for, such as tracking exertion, pacing oneself, or motivating rest [29, 59, 72, 132]. Metrics ostensibly associated with energy have also been increasingly integrated into commercial self-tracking technologies [2, 3]. Yet, energy as a design material remains an elusive concept for recent research on self-tracking primarily due to its ambiguity and contextual dependencies [116]. It is also unclear if and how energy-related self-tracking practices travel outside the context of chronic conditions management. This paper addresses this gap by examining how the concept of energy is informed by the practices of individuals who also self-track, but not with the intention of managing chronic conditions.

To inspire and situate our research inquiry, we turn to crip theory. Crip theory stems from feminist and queer studies and reclaims the term "crip" in similar ways to the manner that "queer" was reclaimed by the LGBTQI+ community. In design and technology, cripping [57, 62, 64] is a form of resistance against "technoableism" that describes how technologies address disability or illness as something that needs to be cured or fixed. Instead, crip technoscience views disability as intersectional, unstable, fluid, and socially constitutive [51]. We adopt crip theory values and concepts [40, 67, 90] in our study design of a survey and semi-structured interviews on energy and self-tracking practices when these tools are *not* used to manage chronic conditions. This approach allows us to identify how societal norms are embedded in and enacted by self-tracking technologies in relation to the topic of energy tracking, even outside of the stigmatised context of chronic conditions.

As main contribution, this paper presents a novel empirical and qualitative study into energy tracking practices through wearable trackers, where we articulate three energy conceptualisations, and three pacing strategies for individuals not managing chronic conditions. We demonstrate how felt understandings of energy can also be misaligned with self-tracked data for people without chronic



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conditions, revealing how normativities embedded in the design of commercial wearable self-trackers can affect their usefulness as tools to gain a deeper insight into energy fluctuations. The discussion returns to the initial context of dis/ability, situating our findings within existing research on self-tracking technologies adopted during energy-limiting conditions such as long COVID [59]. We contribute with a conceptual scaffolding of energy tracking as emerged in our inquiry that builds explicitly on prior conceptualisations of pacing with self-technologies. Drawing on the frictions that arise from our materials, we then propose three design implications for future research on energy tracking, an overlooked design space with the potential to extend beyond the confines of chronic condition management. We conclude by reflecting on what we can learn from designing *from* the experiences of people with chronic conditions, and the ethical implications of troubling the binaries of health/illness, ability/disability. Ultimately, our crip orientation to energy tracking illuminates design pathways for more inclusive futures for all, where health and wellness are perceived in a spectrum and technologies are made for criptopias [10] which serve a wider variety of health goals and outcomes.

2 Energy in Self-tracking Technologies

2.1 Self-Tracking Technologies and Derived Metrics

Historically, self-tracking systems are built around the collection and display of real-time measurements such as steps, heart rate, or activity tracking. In recent years, commercial wearable self-trackers have introduced longitudinal [41, 137], holistic [115], or derived [16] metrics, modelling bodily rhythms to predict physical and mental exertion and readiness [22]. Examples of these are sleep and stress tracking, strain, and recovery metrics, estimations of maximal oxygen consumption (VO2max), and resting heart rate. They are typically calculated over aggregated heart rate data, often estimated from PPG pulse sensors, skin temperature measured from the palm side of the arm, and accelerometer data over long periods of time [8, 9]. As Bentvelzen and colleagues describe, these derived metrics should not be understood simply as "data sources which need to be presented optimally, but also as complex artefacts with internal structures which can be designed" [16, p.3].

As such, they have been considered proxies for general stress level and how well the body copes with stressors, which can be physical, e.g., strenuous physical activity, or psychological, and so they also serve as proxies for general wellbeing [18, 113]. For the remainder of this paper, we follow Bentvelzen et al. and refer to these metrics simply as *derived metrics* [16]. Prior work studying the correspondence between self-report and wearable measures shows that derived metrics of stress and tiredness do not perfectly match with how individuals experience their bodies and energy levels [120], indicating that more research is needed to understand how interaction with metrics should be designed to better support individual wellbeing decisions.

2.2 Energy Tracking in HCI

Research in the field of Human-Computer Interaction (HCI) on energy and alertness typically has addressed tracking the drowsiness of people driving vehicles [65], cognitive alertness in various settings [36, 131], as well as technologies that help individuals fall asleep and wake up [71]. Limited prior studies on modelling bodily rhythms [22], consolidating information on awake-sleep cycles [9], menstrual cycles [78], fitness training-resting cycles [31] have conceptualised energy as a finite health resource or as an element of ecological energy states [74, 116]. One notable exception to this gap in research is Searle et al.'s recent study that found that energy definitions are diverse and amorphous and that this should be supported by the design of energy tracking technologies in order to help users reflect on their emotions and activities in daily life [116]. Our research differs from this study by specifically studying the phenomenon of energy tracking in dialogue with prior research on the experiences of people with chronic conditions.

Despite a general lack of research on energy in HCI, there is an increasing awareness that energy tracking practices are particularly significant for people with chronic conditions [74] and particularly energy-limiting conditions (ELCs) [29, 47]. ELC is a term conceptualised by disability researchers and communities to describe conditions marked by significant energy constraints. These conditions typically include long COVID and ME/CFS, yet other chronic illnesses and conditions fall under the umbrella of ELCs. For example, fibromyalgia, multiple sclerosis, Parkinson's, and conditions such as concussion, neurodiversity, stress, burn-out, and depression [47]. A major symptom that can be experienced in connection with ELC is post-exertion malaise (PEM). PEM describes a worsening of symptoms (sometimes referred to as a "crash") as a result of physical or mental overexertion [52]. In an effort to manage limited energy levels, people with an ELC limit activity and balance exertion with periods of rest. This practice is known as "pacing" [42, 89, 110]. Learning how to pace can be a challenge, as each person has to learn their personal limits, and, due to the dynamic and changeable quality of chronic conditions, these limits can fluctuate day to day [83, 102].

The unpredictable and under-understood nature of ELCs leads to a wide range of self-management strategies [76], using wearable devices such as Fitbits, Garmin watches, and Apple watches, and apps like MyFitnessPal, Google Fit, and Apple Health. Previous research [27, 29, 55, 59] has highlighted the benefits of these technologies in aiding users to gain a sense of control over their under-studied condition [94] through "objective" data, and in helping structure pacing by providing retrospective data on the physiological effects of particular activities on their wellbeing. This means users can plan future activities based on how much energy they will "cost" [59]. However, since fitness tracking devices are designed with the intention to encourage the user to "do more", this may clash with the usages aimed at "doing less" [59]. We return to this research in our discussion, where we will draw comparisons and conflicts between our findings and the tracking practices of people with ELC.

3 Crip Methodologies

Crip theory emerged from the field of critical disability studies and draws on feminist and queer theory to interrogate the political and societal norms surrounding bodies and abilities [88]. Crip theorists resist the historical medical models of disability as an inherent trait to be improved through medical interventions, and instead

interrogate the social and political factors that, often negatively, inform the experience of disability [67]. Aimi Hamraie and Kelly Fritsch's "Crip Technoscience Manifesto" introduces the concept of *"crip technoscience"* which explores the intersection of technology, science, and politics of the body through crip theory [51]. Adopting a crip perspective in design—in other words, cripping design—is an active form of resistance against the idea of "technoableism" which represents technological efforts to solve disability [57, 119].

Crip approaches in HCI research reject essentialist meanings of embodiment [125], productivity [58, 63, 64, 123], tools [108], and the design of biosensing technologies [61]. Specifically in relation to chronic conditions, recent studies have adopted crip theories to co-design technologies for ELCs [57], to articulate interdependent relationships with health technology [39], and to explore the temporal effects of chronic conditions [19, 20, 118]. Crip theory has also been applied outside the context of chronic illness to critique and rethink accessibility and interactions with technologies whilst attending to intimate, temporal and fluid aspects of menstrual pain and discomfort [99, 100]. We extend this HCI literature by adopting crip theory to inform our methodology to study energy self-tracking practices, outside of managing chronic conditions. We do so to inquire into the normativities themselves—as they may be felt by a larger population than energy self-tracking literature centred on chronic conditions suggests, as well as to build evidence towards new ways of designing wearable trackers that all can benefit from.

3.1 Troubling the Binary of Dis/Ability

Crip theories critique and challenge normativities with a vision of care and support as a pleasure and not a chore, and push the boundaries of disability studies to the point that perhaps even disability can become the rule, and normalcy the exception [7]. McRuer argues that *"able-bodiedness [...] still largely masquerades as a nonidentity, as the natural order of things"*, and proposes the term "compulsory able-bodiedness" in reference to "compulsory heterosexuality" [88]. We explore whether this totalising effect of "compulsory able-bodiedness" in the context of self-tracking has the effect of erasing nuances that might otherwise have allowed us to identify similarities between ability and disability. Troubling these binaries is important for crip endeavours to resist the innate universality of being healthy as an identity [134].

To trouble these binary distinctions, Goodley draws attention to the dominant cultural imaginaries around disability/ability that are often constituted as distinct polarised entities and proposes the terminological shift to "dis/ability" that recognises how disability and ability are always reliant upon one another. In order to think of disability, we must pull into the foreground the entity that is ability, and vice versa [45]. The term "dis/ability" therefore reflects the liminal space between ability and disability, interwoven with the in-between politics of bodies and identities [44]. Similarly, Ruth Pinder [103] and Alison Kafer [67] previously reflected on the fallacy of being either "sick" or "fit" independently of the emergence of the concept of dis/ability. More recently, Thorneycroft argued that attending to ableness in crip inquiries, what they call "crip abledness", acknowledges that, similarly to disability, ability can be performed in varying ways, and therefore turning to crip knowledges and epistemologies helps researchers not only better understand disability itself but also interrogate what we know, "around" and "through", disability [134].

Subsequently, we approached self-tracking experiences relevant to energy in a way that allows for a variety of "crip abledness", i.e., ability performed in different ways during the normative self-tracking of people not managing chronic conditions. Our methodology de-emphasises the binary of dis/ability and encourages a movement towards crip futures for everyone, mainly through taking from the practices of people with chronic conditions as an analytical lens, and through our recruitment strategy, described in section 4.2.

Inspired by crip approaches that resist replicating norms and assumptions in health ideals through our research methods, we chose to let the concept of energy remain undefined through the study [57, 82]. We instead welcomed individuals to contribute their meanings and practices. For this purpose, there were two phases in the study. The first phase involved a survey aimed at understanding *which technologies and data types were being used in relation to energy tracking and management*. The second phase, the interview study, intended to explore *how energy is perceived in relation to self-tracking technologies but is also inevitably affected by their support systems, physical accessibility, and embodied experience, social obligations, and the demands of a world built for able bodies*. Therefore, the analysis of our interview data also required a similar crip approach, examining the interplay among tracking and metrics expectations, participants' embodied experiences, their personal health expectations, and in particular, how social and cultural norms informed these factors [63].

4 Method

4.1 Study Overview

The study included an online survey and optional follow-up semi-structured interviews centred on participants' interactions with metrics from wearable devices in relation to their understanding of energy levels.

4.2 Recruitment

We received pre-approval from the Swedish National Ethics Board. We recruited individuals who were 18+ years old and were using wearable trackers. In order to differentiate our research from prior studies on the use of energy tracking to manage chronic conditions, our recruitment process ensured that the purpose of tracking amongst participants was not currently anchored to the management and monitoring of a chronic condition or disability. However, we did not collect any information regarding participants' prior health state or disabilities, and we left participants' definition of chronic conditions open to interpretation. If participants contacted us to ask for more information, we confirmed that they were eligible if they were not currently using a tracking technology to manage a chronic condition. As we discuss later in the paper, this approach allowed for findings relating to how our participants' use of energy tracking and understanding of their energy levels were informed by prior learnt experiences of chronic conditions and/or disability. The first author designed advertisement materials describing the study and survey. Survey advertisement materials were distributed on authors' social media channels, posted around university campuses,

and posted on Reddit and Facebook communities of fitness and health trackers and tracking enthusiasts.

4.2.1 Participants. In total, 50 individuals completed the survey (Median Age=34.5), 26 of whom indicated interest in conducting a follow-up study. Eventually, 15 respondents participated in the interview part of the study 1. We anonymously collected information on individuals' age, country of residence, gender, and household type to scope the population of interest. Survey respondents were mainly from European countries (n=37), the United States (n=11), and India (n=1). Whilst most interview participants were residents of Nordic countries (n=13) and the United States (n=2), many of those participants had other cultural and ethnic origins. The survey collected responses from 30 females, 19 males, and 1 non-binary person, while the interviews were conducted with 14 females and 1 male.

The gender imbalance amongst interviewees led us to return to the survey ratings to delineate potential connections between gender and metrics' usefulness to energy understandings. Female respondents were more prone to rate the impact of self-tracking metrics on their understanding of energy higher than male respondents. Therefore, we speculate that female participants might have been more motivated to continue participating in research on a topic they found valuable. However, we cannot be certain of the causes for this disparity since we did not ask participants about their decision to participate or not.

4.3 Study Design

4.3.1 Survey. The survey was deployed in Microsoft Forms, and responses were collected anonymously. In case respondents wished to be contacted to schedule an interview, they could then —and only then— share their email address. Respondents had the freedom to withdraw interest at any time, which meant that they wouldn't be contacted again.

The survey firstly collected the type of wearable tracking devices used, the main tracking motivation, and the self-tracking experience in years. To examine how self-tracking metrics influence energy interpretation, we first created a comprehensive inventory based on categories established in previous research on self-tracking as a proxy for wellbeing [37, 116]. Based on how commercial wearables' manufacturers categorise each of their metrics, we divided the self-tracking metrics of commercial trackers into the following categories: Cardiovascular and Respiratory Metrics, Sleep Metrics, Stress Metrics, Recovery and Resilience Metrics, Menstrual and Pregnancy Insights (if applicable). To account for the diversity of market descriptions, we either utilised broad definitions of a metric's purpose (e.g., Menstrual Day) or distilled specific metrics into overarching themes while providing examples of their commercially available counterparts. Each category used the Likert scale for each metric, rating the usefulness of the metric in understanding and making sense of energy levels and reserves [66]. The survey Likert responses were rated with the range 0-5 (i.e., Not Applicable/NA = 0 and Extremely = 5) to measure each metric's usefulness to energy understandings. If a metric was not featured in respondents' trackers, or not relevant to their case, e.g., menstrual cycle tracking, they were able to select "not applicable". The survey intended to account for the wide range of bodies, energy definitions, and motivations for self-tracking. To achieve this, it allowed respondents to provide reasoning for their ratings with free text boxes and, in the final section, to contribute with other relevant metrics not included in the pre-defined categories.

A descriptive statistics approach was deployed to describe the quantitative data in relation to the textual contributions of respondents [140], where the first author analysed the ratings alongside the answers to open ended questions (please refer to the Supplementary Material) to understand preferences and general uses of metrics as proxies for energy. The responses from the survey were additionally used as part of the interview protocol, as we describe below.

4.3.2 Interviews. Semi-structured interviews lasted on average 38.41 minutes, depending on participants' engagement, and were held online through an encrypted Zoom link. Semi-structured interviews were conducted in English (14) and Greek (1). The interview protocol was divided into two parts and was based on the participant's individual survey responses [28, p.55]. Data engagement interviews are an elicitation technique which incorporates data into the design and development of semi-structured interviews [92]. In a related vein, our protocol was pollinated with survey responses on self-tracking metrics to encourage active, flexible contextual reflection, ensuring personal data practices remain central to the method. First, the participants were prompted to describe their daily rhythms, self-tracking habits, sociocultural settings, and their understandings of energy. Second, the responses to the survey were jointly reviewed to enable discussions on the role and the impact of self-tracking metrics in participants' daily energy understanding assessments, as well as to relate those responses to individual contexts.

The transcribed interviews were analysed following a qualitative content analysis process [75]. The first author selected randomly five interviews and inductively developed an initial set of codes. In a second round, these codes were deductively applied to nine additional interviews, which led to their further refinement. The final, stable codebook was then applied to the entire interview data.

All three authors read through the codes and relevant interview materials in two three-hour sessions. During those sessions, codes were firstly categorised into thematic conceptual areas that describe how energy tracking is done. Five themes resulted from this analysis: 1) *Energy as a Concept*, 2) *Energy Understandings in Interactions with Self-Tracking Biosensing Metrics*, 3) *Designing for Personal Energy Meanings*, 4) *Bodily Knowledge in relation to Energy Framings*, and *5) Normativity and Misalignment in Energy Conceptions*. Next, all authors revisited the analysis refining, eliminating and merging themes. This final analytical step resulted in three overarching themes that relate to 1) Energy Conceptions, 2) Energy Management Strategies, and 3) Frictions and Normativities, presented in Section 6.

5 Survey Results: Energy Tracking Metrics and Proxies

Survey participants were prompted to choose or specify their health and fitness tracking devices, along with their tracking motivation and their years of experience tracking. 13 out of the 50 individuals indicated simultaneous usage of several self-trackers. 9 respondents

Participants	Age	Country of Residence	Type of Wearable Tracker	Years of Tracking Experience
P01	37	Sweden	FitBit	3-5 years
P02	35	Sweden	Samsung Galaxy Watch	less than 1 year
P03	35	Sweden	Garmin Watch	more than 5 years
P04	44	Sweden	FitBit	more than 5 years
P05	29	Sweden	Oura Ring	less than 1 year
P06	35	Sweden	Garmin Watch	less than 1 year
P07	31	Sweden	Oura Ring	1-2 years
P08	33	United States	Apple Watch	3-5 years
P09	22	Sweden	Garmin Watch and Polar	3-5 years
P10	65	United States	Ultrahuman Ring Air	less than 1 year
P11	43	Sweden	FitBit	more than 5 years
P12	27	Denmark	WHOOP Strap and Apple Watch	1-2 years
P13	31	Sweden	Samsung Galaxy Watch	less than 1 year
P14	25	Denmark	Garmin Watch	1-2 years
P15	27	Sweden	Xiaomi Redmi Watch 3 Active	3-5 years

Table 1: Table describing interview participants' code, age, country of residence at the moment of the study, types of technology used and experience tracking.



Figure 1: Survey responses regarding usefulness of various metrics to energy understandings. Energy-proxy metrics in the figure (e.g. metrics relevant to energy, bodily Exertion, recovery and restorative States, and training Load and Exercise Exertion) were presented to participants with the respective examples from commercial trackers.

identified fitness training to be their primary motivation for usage of tracker, with the rest of the respondents describing, as primary motivations, stress, sleep, and overall health objectives such as

monitoring daily activity and movement levels, fatigue monitoring, reflecting on various health metrics with one self-tracker, and increasing personal accountability.

An overview of the ratings of usefulness of types of self-tracking metrics associated with the process of understanding energy levels and reserves is presented in Figure 1. Sleep-relevant metrics were highly ranked, and particularly the sleep score that assesses the overall quality of sleep. The category of sleep metrics was found, on average, irrelevant to energy understandings by only 6.75% of respondents, contrasting with 54% (n=27) of participants rating these metrics as helpful or extremely helpful. Heart rate-based metrics such as the resting heart rate (RHR) and heart rate variability (HRV) were also highly rated in terms of usefulness and relevance to understanding energy levels. Body temperature was the least useful in this category of metrics in relation to energy understandings.

Derived metrics associated with bodily exertion and resilience were the least popular amongst all participants. 24% (n=12) of the responses indicated that they found the metrics used by manufacturers as proxies for energy levels and readiness, such as Garmin's Body Battery, Oura's Readiness, Fitbit's Readiness, WHOOP's Recovery Score or Polar's HRV Status, not useful, or slightly useful. Metrics associated with bodily exertion (e.g., DayStrain (WHOOP)) were rated as not applicable to energy understandings by 68% of responses (n=34). However, metrics given as proxies for daily recovery were selected as relevant for energy understandings by many of our participants, with 22% (n=11) of responses finding them very useful, and 4% (n=2) extremely useful. Lastly, metrics that describe training load and exercise exertion were rated as very (20%, n=10) or extremely (14%,n=7) useful.

Respondents who found the tracking of the menstrual cycle applicable indicated the phase, as well as the day of the menstrual cycle as significant to their daily understandings of their energy levels, connecting the menstrual cycle to experiences of fatigue and low resilience. Several survey respondents clarified that they do not necessarily track menstrual cycle information with the same wearable tracking device, preferring for example to track body temperature and mucus changes in conjunction to other metrics to make decisions based on their energy levels, and ability to recover from exercise.

Participants also reported using metrics not explicitly mentioned in the study but that contribute to their daily energy understandings. We collected 19 responses that named consistency in various types of movement-based metrics (i.e., time standing, calories burned, steps, or fitness and performance markers) as a significant identifier of energy levels. Some responses noted the importance of metrics from non-wearable tracking sources, for example, the Flo menstrual tracking app that offers daily energy prediction based on the menstrual cycle phase [1]. Other respondents described how they complement their tracking with self-reported information that they have experientially identified as relevant to their energy fluctuations, for example, the intensity and balance of social activities, daily tasks, and restorative time.

The survey demonstrated the relevance of energy tracking for our participants, showing that metrics from tracking technologies are appropriated for this aim. With the interviews, we describe how metrics are enrolled in strategies and conceptualizations of energy.

6 Interview results

Here we present the three themes resulting from the interview analysis: 1) Energy Conceptions, where we describe how participants conceptualise their felt energy, 2) Energy Management Strategies, and 3) Frictions and Normativities, where we describe how our participants experienced misalignments between their felt experiences of energy and the metrics.

6.1 Energy Conceptions

6.1.1 Energy as a Measure of Wellness. Heart rate variability (HRV) was notably regarded as an indicator of energy status in relation to participants' general wellbeing. Several participants (01, 03, 09, 10, 11, 14) explained how their overnight HRV could be a sign of sickness or stress, which would explain and confirm lower energy levels. As described by P09, *"So for me, [HRV] usually drops before I get sick. So then I might know if I need more rest or if I need to like calm down. And often times you have the intuition that it's been a stressful period [...] I feel like the heart rate variability can be kind of a confirmation of how you've been feeling"* (P09). Similarly, most participants identified Resting Heart Rate (RHR) as correlating with fluctuations in their health and wellbeing and their ability to recover after a hard workout. Participants also noted that their experience interacting with respiratory data from their tracker (e.g., VO2 max, FTP) correlated with low energy levels (P06 and P14). P14 also described how these changes were due to illness, but might also be due to *not* having worked out as much as usual.

6.1.2 Energy as a Reserve. Daily engagements with activity and exertion-related metrics contributed to personal interpretations of energy, particularly in relation to personal thresholds for physical exertion (P02, P07, P13, P15). The numerical representation of physical activity and movement (e.g., heart rate, stress score, calories burned, steps completed, time being active) made energy expenditures tangible and informed participants of how quickly their energy reserves were being used up. Participants described how running out of energy would have negative effects on recovery time and health in general. Energy was also seen as something finite that decreased over a lifetime by P08, *"Like I feel like I have less energy than I did when I was a teenager or in my early 20s. So I could imagine what I'm trying to do now is to try to preserve energy for the future that I want to be able to continue to have energy as I get older".*

6.1.3 Energy as Cyclical. Amongst the participants who ovulate and menstruate, there were two distinct perspectives on how menstruation and ovulation relate to energy definitions. Participants 01, 02, 03, 04, 07, 11, 13, and 15 highlighted that menstrual and ovulation cycle insights and metrics informed their understanding of their energy levels. The other cohort of menstruating participants did not find this information particularly relevant in regards to their energy levels. The hormonal contraception was the main cause due to the effects on altering menstrual cycles and suppressing menstrual bleeding. Participants who did find this metric useful described being able to correlate low energy levels with menstrual phases before worrying about other factors. As described by P13, *"Energy-wise, I have a few days where I feel more exhausted and just more prone to headache and cramps and just being physically a little beside myself. I think if I can explain that through the menstrual cycle,*

then I can accept that I feel a little bad, but it's okay because it's just because of the hormones". Whilst significant, keeping track of their menstrual cycle with self-tracking technologies is difficult due to their reduced and irregular periods of menstrual bleeding.

6.2 Energy Management Strategies Informed by Self-Tracking

We found a range of strategies used by our participants to either influence or augment their energy levels, or to allow them to make the most of their energy levels in order to conduct specific activities.

6.2.1 Optimising Energy in Order to Do More. Cardiovascular data such as heart rate (HR), heart rate variability (HRV), and resting heart rate (RHR) were the most commonly discussed metrics in relation to energy levels and expenditure during physical activity (P02, P06, P11, P14). These discussions typically centred around how to use data in pursuit of achieving fitness goals, particularly in making the optimal use of energy reserves through using heart rate "zones". These zones are different heart rate values and how they correlate to different types of activities and wellness goals, including avoiding over-exertion that would delay recovery and deplete their energy reserves in the future (P08, P02). As P11 described, monitoring allowed them to keep *"control"* over their heart rate, keeping it low when running through using slow breathing exercises. As well as being meditative, this allowed them *"more time on the treadmill without getting tired."* Maintaining a lower heart rate during physical exertion allowed for longer periods of activity overall and less recovery time.

Other examples of these types of strategies included optimising sleep to optimise energy levels the following day: *"the sleep score [...] is directly related to caffeine and directly related to a late meal. So I try, I don't have late meals and I don't have caffeine after lunch anymore"* (P10). Participants described valuing consistency in sleep data in relation to reducing fluctuations in energy overall during the daytime. As described by P08, *"By getting more consistent sleep, I'm helping myself through those difficult times. I like seeing the consistency of my sleep because I know that can only be helping my energy levels."* (P08). Participants also described the negative effects of a lack of physical activity on their energy levels (P13, P14, P15). They described how data relating to how sedentary they were during the day; *"I might be a little bit more sleepy or something. (If) I made that change to be standing for a little bit longer, it will wake me up a little bit more and I'll have more energy to do things with more alertness"* (P15).

6.2.2 Utilising Physiological Fluctuations in Order to Optimise Energy. Participants noted the beneficial impact of using metrics from their wearable trackers to explore how they could align when they exerted energy with their bodily rhythms (P11, P12, P05). Self-tracking trends and insights allowed participants to harness the energy fluctuations of their menstrual cycle or circadian rhythm and adjust their activities, habits, and diets in ways that make sense to them (P10, P02, P07). P7 described how understanding the menstrual cycle could allow her to increase activity; *"I just find it so useful to know where I am in my cycle for many things. I think one of them is energy. I have really big shifts when I go into my luteal phase, so right after ovulation, and I feel that in my body, and it's like really good to just keep track for me and to know because I guess if I can, I try to do different kind of things during different phases of the cycle".*

6.2.3 Validating Rest and Doing Less. The previous sections illustrate a cluster of tactics for how self-tracked data can be used to optimise energy levels in order to *do more*. However, we also found examples of the use of strategies for rest and *doing less* amongst our participants. Heart rate and heart-rate derived metrics (i.e., HRV, stress levels, and RHR) acted as an estimation of participants' energy capabilities for the day, serving as a prescription to slow down and/or recover (P02, P03, P09, P10, P13). Participant 02 expressed her strategy of monitoring heart rate, especially during periods of high stress to *"take the decision if I am reaching my limit and I should slow down"*. Data was used to excuse the prioritisation of the personal strategies of recovery and to reduce feelings of guilt and uncertainty (P03, P09, P11). As P03 explained, her consistently low HRV values forced her to do *"a bit less because it seems that my body is kind of asking for it indirectly."*

Aligned with the findings of the survey, most participants' perceptions of their energy levels were influenced by sleep metrics calculated by their wearable tracking devices (P01, P08, P09, P15). Participants felt that the range of sleep metrics often correlated with the quality of their recovery from exertion the previous day. Consequently, sleep metrics indicated energy reserves for the day ahead and dictated the types of activities they could do sustainably; *"I definitely know that if I see, for example, that I've not slept well, then I know that I'm not going to have a high energy day"* (P7). Similarly, this data could be used to excuse lower activity: *"If [the score is] low, I feel less energetic but less guilty about taking a nap after work."* (P05).

The interactions with metrics related to exertion and recovery (i.e., readiness, strain, and recovery scores) were similarly described to act as a compass for identifying energy limits and restrictions and choosing between intensity of engagement with tasks and activities (P05, P06, P03, P12, P13). For example, readiness scores helped P05 to identify the straining dimensions of travelling whilst supporting her re-adjustment to old routines by fostering post-activity care and rest. *"This was the first time I used the metrics after such a trip, and it made me realize how much I need to plan two or three days of rest when I come back. It's not strange for me to feel tired after a long trip because the metrics show that."* (P05). Similarly, stress metrics could also be used to predict and manage energy expenditure to prioritise rest; *"I can also see in which situations I see the stress level going up and that maybe it gives me a bit of a warning sign of hey maybe you need to rest."* (P05). P12 found that her "recovery level" score better reflected her bodily fluctuations than her previous use of calories spent to track her activity levels: *"I am now trying to adjust my activities to the 'recovery level' because I used to (use) this 500 kilocalories every day (metric) and now I'm realizing that my body is not a robot [...], it's alive and it's different every day. [...], having this recovery score allows me to see that if I feel tired and exhausted, maybe I can do less and spend more activities on, to do some more recovering activities."* (P12). Conversely, participants also reported choosing to ignore "bad" sleep and stress scores in order to conduct activities that improved their quality of life, such as travelling (P05) or socialising (P05, P06).

6.3 Frictions and Normativities Relating to Self-Tracking and Energy

All participants shared frictions and misalignments between self-tracked metrics and their own felt experiences of energy.

6.3.1 Mis-Interpreting the Body and its Signals . Many of the most recent versions of wearable trackers provide energy-proxy metrics that quantify daily dynamic recovery. These metrics are offered with the aim of helping users assess their capabilities for the day and/or monitor their strain and recovery after working out. However, these metrics only work when participants are feeling well (P06, P12, P05, P09, P11). As reported by P12: *"[...] I realised that they're a bit toxic because I can, I don't know, do a lot of activities like ride 100 kilometres, and then on the next day they're like, maybe you will move because you are not so active. I was too tired of it and I realised that it doesn't align well with my energy levels."* (P12). Metrics related to resilience may be especially poorly correlated with felt energy during temporarily unwellness: *"when I'm sick, my readiness score is often higher than it really feels. Recently, I was on a train and got really motion sick. I felt nauseated, but my readiness score was super high because I spent five hours sitting down. It was telling me to be active, but I just wanted to throw up"* (P05). Inversely, P6 reported feeling highly energised and even positively *"hyperactive"* in the evenings, but her tracking device often advised her to reduce activity and go to bed.

Participants described how their understanding of stress is misaligned with how it is measured and reported by their wearable trackers. For example, a day might be labelled as restorative and relaxing based on physical activity metrics, despite users having experienced a high mental load: *"I've been sitting on a desk for 12 hours doing my work and writing and thinking and reading sources and connecting and citing and doing theoretical discussions by myself. [...] I was joking with my [co-living partner]. I was like, oh, apparently today I had three naps while I was at work"* (P06). Conversely, pleasurable activities could be labelled as stressful *"It tells me I've had a stressful day, and I say, 'Yeah, but I've hung out with friends. I was happy and got energy from that.' So, that's very interesting."* (P05). We also found cases where metrics failed to acknowledge significant differences in metrics compared to individuals' baselines. P11 described how she stopped wearing her wristband when sleeping after her tracker interpreted her sleep as average, whilst she was on a journey of overcoming insomnia symptoms. A final type of friction occurs when participants who menstruate experience a disconnection between their felt energy, which is experienced as cyclical and highly correlated with their menstrual cycle, and other metrics. As an example, P3 found her wearable tracker's descriptions of causes of HRV values insufficient, as they tended to focus on tiredness and recovery, and not on their menstrual cycle.

Participants also expressed their frustration with the lack of ability of self-tracking technologies to acknowledge health fluctuations in their calculations. As a result, a low heart rate from birth (P12), or peculiarities in sleeping patterns due to work schedules (P09, P11) received negative scores and feedback from tracking devices without letting individuals contribute to setting their own baseline. Rather than being dissuaded from tracking or being persuaded to change habits, P12 and P09 described developing their own strategies of interpreting energy in relation to their sleep score and their resting heart rate. They both chose to ignore the rankings and pay closer attention to their own baselines informed by their previous experiences of monitoring such metrics.

6.3.2 Knowledges Drawn from Fluctuating and Unwell Bodies . We found examples where either previous illness, physiological processes, or other people's experience of chronic illness were useful, and even required, to understand the nuances of data related to energy levels. Despite P04's hysterectomy and lack of tangible ovulation signs, she described her ability to adjust her self-tracking practices to the embodied signs that approximately mark her ovulation day. This allowed her to continue to track her menstrual cycle, and therefore her cyclical energy levels, despite not fitting the normative definitions of menstrual events necessary for menstrual tracking (e.g., menstruation). Likewise, past experiences of endometriosis flare-ups were significant factors in P02's knowledge in interpreting cyclical menstrual symptoms and have previously brought awareness to their associations with energy fluxes. P01 described how her husband's expertise due to chronic illness informed her own tracking and energy management. P01's tracking motivation is to maintain an active lifestyle, and whose daily energy interpretations are based primarily on her sleep data. Her husband has had long COVID for the last three years. Together, they engage in a daily ritual of data sharing where her husband uses his experience of managing his limited energy to support her in developing an understanding of her own energy capacity for the day. Participants lastly shared how they understand their energy in relation to their menstrual/ovulation (P02, P04, P07, P11) and sleep cycle (P04, P11), but also within a larger scheme of fluctuation influenced by the changing seasons and variations in daylight and even the waxing and waning of the moon. However, based on their observations current metrics generally make no space for those factors to be accounted for.

7 Discussion

In our study, participants conceptualised energy in various ways: as cyclical (accompanying temporal bodily rhythms like the menstrual cycle), as reserve (a finite resource to be managed and replenished), or as sign of wellness (a signal of illness or health). These conceptions are not mutually exclusive but rather overlapping, and are responded to by energy management strategies to care for the body's needs or temporal fluctuations. Some of these echo pacing strategies found in energy-limiting conditions tracking literature [55], as we discuss below. We also found traces of chronic conditions management strategies in participants (P01, P02, P04) who were no longer strictly 'ill' or were living with others who manage chronic conditions. This aligns with the crip theory critique of a curative future [67] — the expectation that one must strive for a state completely free of disability. However, rather than accepting a binary of 'cure' versus 'no recovery', our study shows that these practices persist even when individuals do not track for chronic conditions management.

We now discuss the findings of our study in relation to previously described practices of energy self-tracking in chronic conditions [29, 55, 101, 114]. We further address frictions relating to the self-tracking and energy, and reflect on our adoption of crip theory.

7.1 Dis/ability in Energy-Tracking: Pacing Practices beyond the Management of Chronic Conditions

The practice of deriving estimations of energy through data (e.g., HRV, RHR, sleep score, menstrual cycle phase) was used broadly by survey respondents and interview participants to indicate energy reserves, make mico-interventions to their energy states and reflect on restorative and exertive activities. Despite variations in preferences for metrics, participants described how metrics often act as an "excuse" to rest and prioritise care for the body and mind. Conversely, participants also described the tactical prioritisation of social and psychological wellbeing over metrics. This tactical interaction with metrics resembles McDonnell et al.'s concept of "consequence calculus" used to manage chronic conditions to guide decision-making based on which activities will provide psychological benefits, in spite of the possible negative physiological effects [83]. Even outside the management of chronic conditions, we observe the persistence of societal perspectives on rest as being unproductive [30, 53]. It seems that it is not only people with ELCs who require metrics as a 'diagnosis' as an excuse to slow down.

Our study extends aspects of the three modes of pacing defined by Homewood et al., in their study on managing energy-limiting conditions (ELCs) using fitness trackers [55, 59]: pacing based on a diagnosis, preventative pacing, and real-time pacing. We relate our findings to each pacing mode, and propose a conceptual scaffolding for energy tracking beyond the management of chronic conditions, also illustrated in figures 2, 3, and 4.

7.1.1 From Pacing Based on a Diagnosis to Metrics as Indicators of Energy Reserves. Homewood [55] identifies the use of data as a diagnosis to inform pacing decisions. For people with ELCs, self-tracked data gives a more accurate picture of the size of daily energy reserves than more generalised pacing methods [29, 55, 59, 101, 114]. Derived metrics such as sleep data, "body battery", and "readiness score" are seen as particularly useful by people with ELC to determine energy allocation for a particular day.

Our participants described using data equivalently as a "diagnosis" of energy, guiding decisions about which activities are appropriate. The most common examples were related to sleep metrics, where a bad night's sleep would affect their expectations of the following day. We also found stress metrics to have a similar effect. In a more literal way, and similarly to people with ELC, we found how bouts of illness could be "diagnosed" through changes in their tracked data. Longer-term, cyclical, physiological fluctuations visible in data, such as the menstrual cycle and circadian rhythm, could also act as a diagnostic tool. These diagnoses helped recognise a felt sense of low energy and, most importantly, acted as a validation for participants' decision to rest or slow down.

7.1.2 From Real-Time Pacing to Energy Micro-Interventions. Real-time pacing refers to the use of data in real-time to inform short-term pacing decisions. For people with ELCs, this meant making pacing decisions based on the number of steps they had already walked for the day and slowing their walking speed down if they found their heart-rate went above a self-determined metric. Tracked data are also used in the moment to dictate when to restrict exertion to "recharge" their energy reserves [55, 59]. In our findings, we encountered examples of the use of real-time data such as heart-rate, activity score, and step count to inform decisions in the near future relevant to personal energy tactics. In contrast to people who pace ELC, our survey and interview participants did not describe dedicated derived metrics, characterised by the manufacturers as proxies to energy levels or readiness, to be useful [34]. Instead, the examples that we encountered included the use of real-time data within the day and/or whilst working out to ensure that personal thresholds of bodily movement and sedentary time were not surpassed. In this case, standing up and moving around after a long period sitting working to combat fatigue and increase felt energy levels in the short-term. Here, self-tracking technologies are used to prompt more, rather than less, physical exertion to combat mental exertion.

7.1.3 From Preventative Pacing to Retrospective Learning about Restoration or Exertion. Preventative pacing involves consulting retrospective data to improve aspects of longer-term, future health. Examples of preventative pacing by people with ELC involve routinisation to understand the energy "cost" of activities [29, 59]. This knowledge is then applied to create a structure and routine that gives better control over the energy distribution, avoiding repeated cases of overexertion that would slow down future recovery.

Our participants engaged in a type of preventative pacing as they described having learnt from their data in order to maintain more sustainable energy levels overall. This included learning the positive impact of consistent sleep routines, the negative impact of travel on their bodies, and the consequences of going over their heart-rate zones in relation to the time it would take them to recover from physical exertion. Similar to people with ELC, these methods were used to avoid overexertion. However, this tactic was not to explicitly maintain health, but rather to reduce recovery time in the future in order to increase stamina and overall capacity for physical activity in the future. Energy levels were understood as something that could be infinitely increased in the future if healthy practices were employed in the present. However, in contrast to pacing practices in ELC management, low activity levels were also believed to impact negatively future wellness [110]. This reflects a foundational difference between people tracking to manage chronic conditions and our participants. In the context of chronic conditions, energy is seen as a finite resource to be preserved, and the negative consequences of overexertion outweigh the potential benefits of increasing future capacity through more traditional approaches to optimising fitness [59, 130].

Learning to navigate energy as a finite, idiosyncratic resource is central to tracking in both ELC and non-ELC contexts, even with important differences discussed earlier. Addressing these contexts from a crip perspective requires balancing immediate protection against future exertion. Ultimately, the trackers used by our participants in this study all lack features that allow users to map the restorative or draining nature of their daily practices, privileging instead the achievement of fitness goals. Incorporating features that help participants learn how bodies respond to different activities over periods of time could potentially support users in adapting to the fluctuating realities of the body. These are practices that our

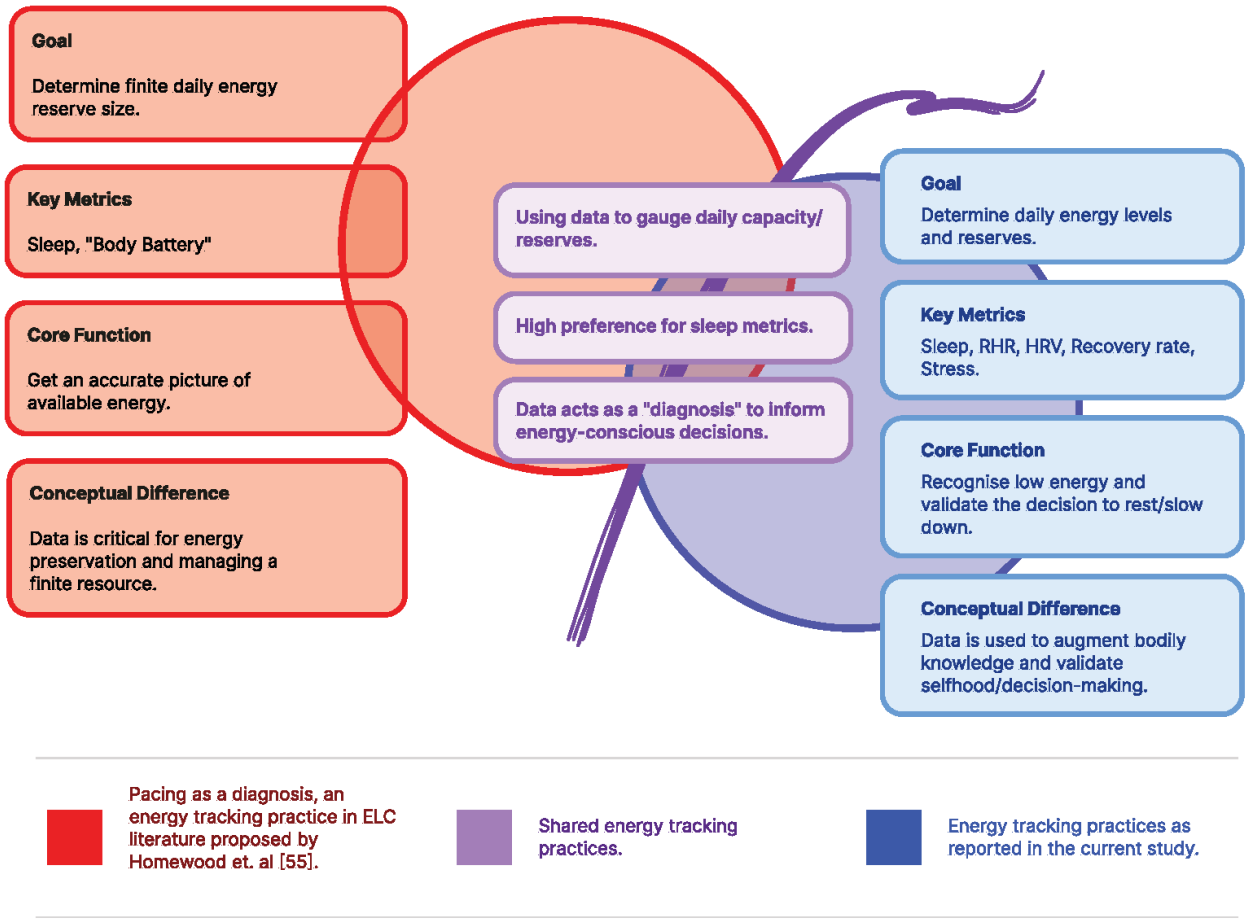


Figure 2: Comparison between pacing as a diagnosis-based energy tracking practice [55] and the practice of metrics as indicators of energy reserves as framed in our study.

participants are already attempting to perform, albeit by appropriating metrics from fitness trackers rather than having these features specifically designed for these purposes.

The introduction of features that support pacing outside of chronic illness management is likely to encounter frictions. In the following section, we discuss frictions that we identified in our study and we propose design pathways to mitigate them.

7.2 Addressing Frictions in Self-Tracking and Energy Through Design

Whilst we would expect to see fewer frictions than in unanticipated uses of self-tracking to manage chronic conditions, it is notable that frictions also exist in our study. These frictions, and the societal norms they represent, suggest research gaps in the design space of personal energy self-tracking.

7.2.1 Friction 1: Normativities in Baselines of Metrics that Act as Proxies for Energy. Self-tracking and biosensing technologies have been previously criticised for their tendency to misinterpret the plurality of bodies and bodily abilities [13, 33]. Similar to the previously mentioned studies on ELC, our participants also described normativities and frictions in interactions with metrics, resulting in participants being recommended to move when they were sick or mentally overworked. Physical activity detection over-relies on biosensing and algorithmic processing [23] and can even restrict user's abilities through not supporting activities important to them, such as going for a walk [139]. We found that sleep and stress scores did not align with participants' felt sense of energy as they seek to impose pre-defined, normative, sleep schedules [69]. When metrics from self-tracking devices did not fit bodily experiences, more than one participant described how they ignored the rankings and instead internally decided on baselines informed by their

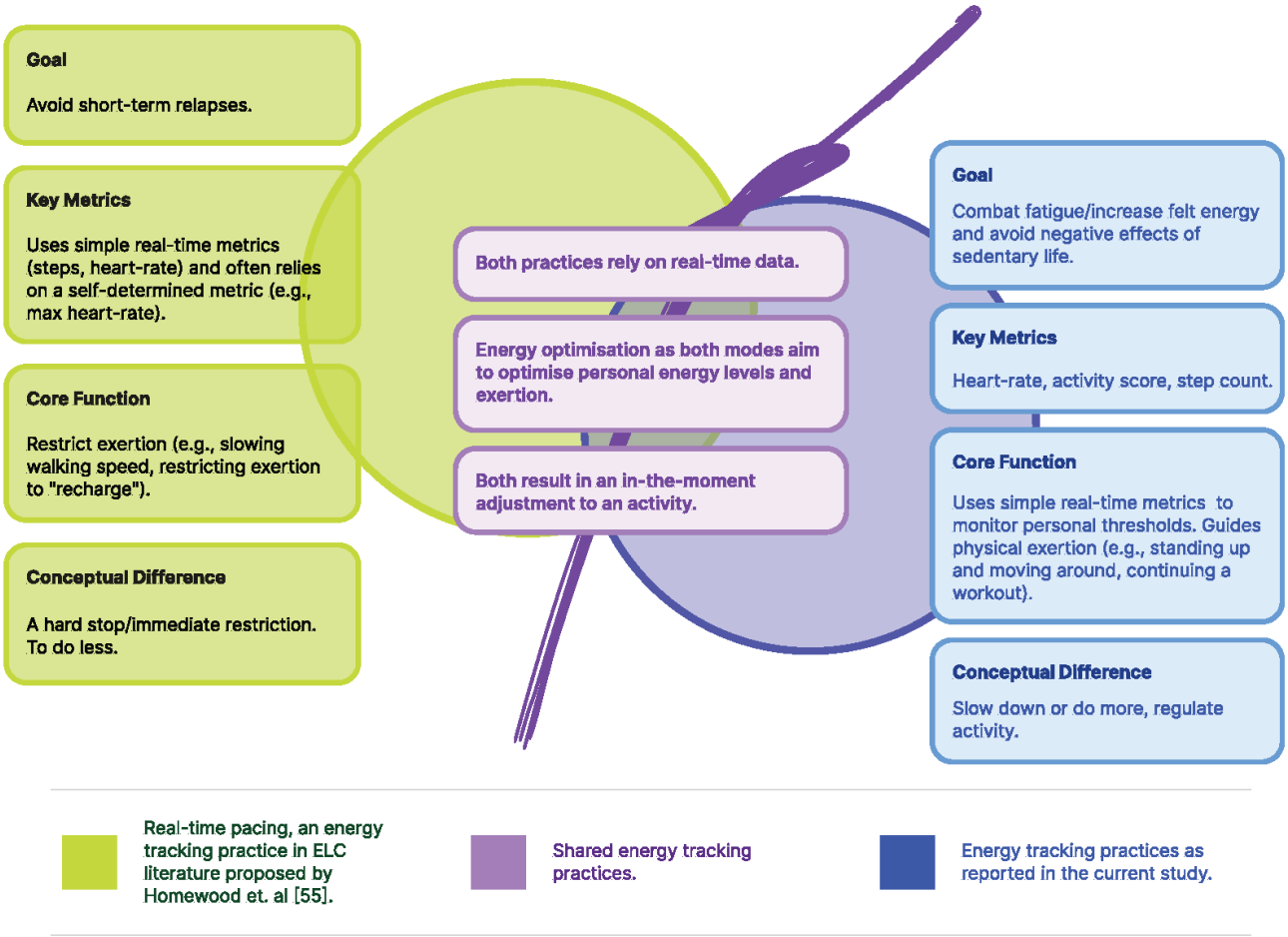


Figure 3: Comparison between the energy tracking practice of real-time pacing [55] and the practice of energy micro-interventions as framed in our study.

previous experiences of monitoring such metrics, sometimes even stopping tracking entirely. These examples demonstrate normative expectations for behaviour, wellbeing, and unmarked healthy bodies [56, 124, 125, 135] embedded in self-tracking metrics which also manifest in energy tracking.

Design Implication 1: To address this friction, we suggest that future interaction design around energy tracking should incorporate elements of interactional calibration, and transparency.

In chronic illness tracking literature, *customisation of data representation* is often given as a solution to mis-interpretation [29, 59, 114]. As proposed by Van den Berg, individuals should be able to calibrate their technologies in ways that they account for their bodily specificities [138]. Our study has shown this to be relevant outside of the context of chronic condition management, as examples of mis-interpretation appeared within all the three conceptual modes of energy tracking with wearable technologies. By interactional calibration, we propose that instead of designing for a new universal metric or scale of "energy", future research should explore the question of how tactics of sense-making with metrics can respond to misalignments of baselines. This design consideration is not only about the creation of customisable baselines. Prior research has shown that sense-making practices of self-calibration, and data augmentation transforms data collection and reflection into a collaborative process between users and their devices [68] in which individuals might experiment with alternative ways to integrate subjective knowledge and support their energy tracking practices. Designing for transparency in energy tracking requires a clear demonstration of what the tracker takes into consideration when computing the metric in question, and it is cited in prior literature [15] as a key dimension for designing derived metrics from multiple biosensing measurements. Limitations and uncertainties in

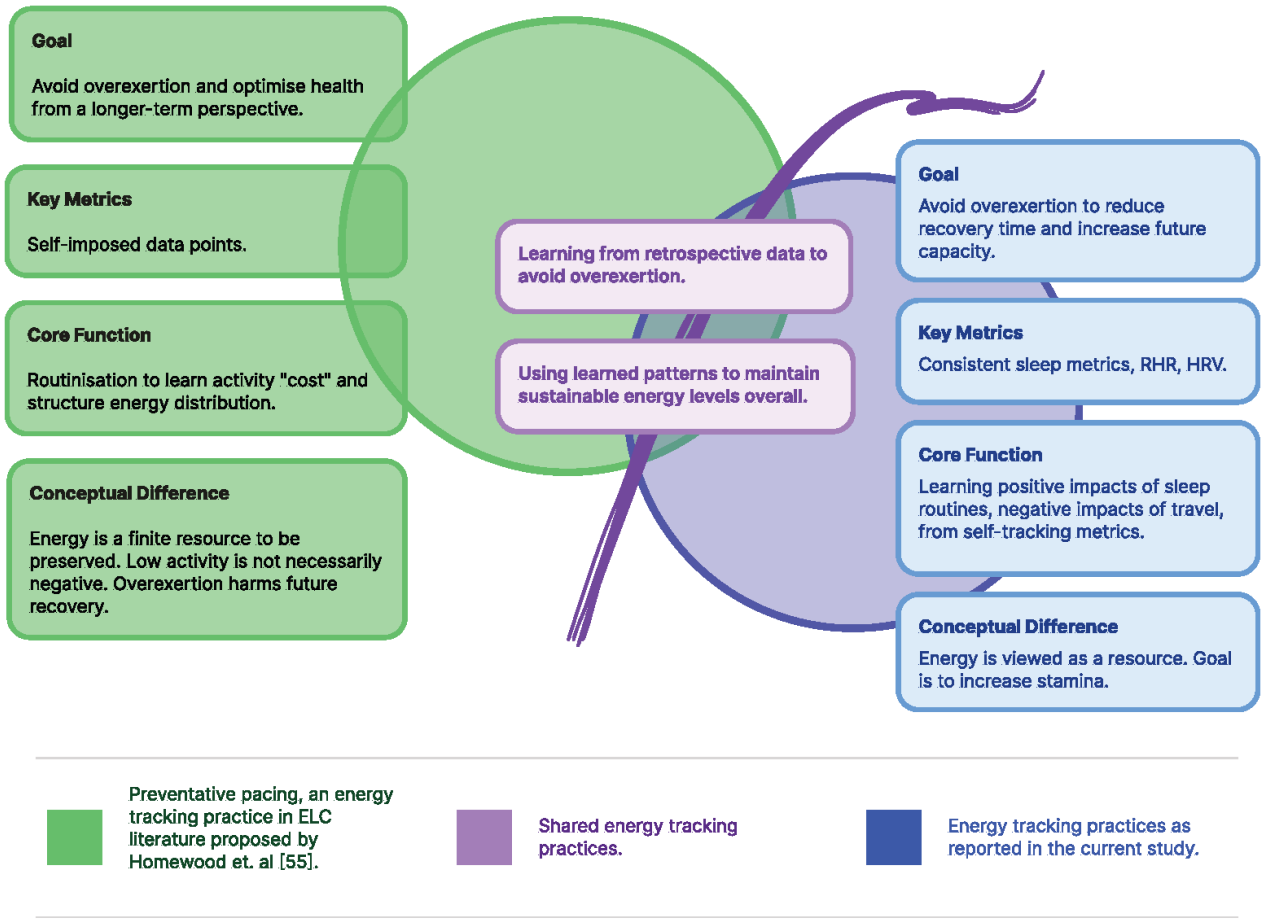


Figure 4: Comparison between the preventative pacing energy tracking practice [55] and the practice of retrospective learning about restoration and exertion as framed in our study.

metrics calculation should be visible to individuals to make visible assumptions [117, 138], and make possible to challenge them. To reduce information load when implementing transparency, prior research has articulated progressive disclosure [127] as a user-centric tactic for the design of transparent systems.

7.2.2 Friction 2: The Cyclicity of Energy Tracking with Acyclic Metrics. Our study showed how the cyclical nature of physiological transitions informs energy understandings, yet remained essentially underserved by metrics offered in fitness and health tracking technologies. Whereas sleep metrics were highlighted as important to interpreting energy in the survey, interviews reported frictions with sleep assessments that seemingly devalued the overall interaction with the tracker. Those cases emerged mainly due to the lack of recognition of larger cyclical patterns such as seasonal changes that, according to participants, should not necessarily need to be conceived as abnormal in the metrics [70]. Participants' experiences of understanding energy with sleep data reflect recent research that shows how sleep behaviours are personal and rhythmic with many cyclic aspects (e.g., circadian rhythm, seasons) and their models are discussed as multidimensional affecting one's health and wellbeing [25, 106].

We also found that felt experiences of energy fluctuations of bodies that ovulate and/or menstruate did not always align with self-tracking technologies. Survey respondents and interview participants highlighted the influence of menstruation on energy levels and expressed their uncertainty regarding whether this dimension was taken into account by the metrics provided by their devices and more specifically sleep, readiness, or stress. Many of our participants resolved this friction by tracking their menstrual cycle in parallel through other non-wearable or non-digital methods, documenting fluctuations in their energy fluctuations there. Others

relied on bodily knowledge acquired through years of menstruation and ovulation. Medical and sports science have reported the overlooked link between hormonal shifts during the menstrual cycle and the ability to sustain and recover from stressors [38, 98]. Medical studies on wearables, biosensors, and the menstrual cycle argue that for certain derived metrics (i.e., HRV and RHR), cyclic characteristics are involved, and others (i.e., sleep, stress score) are not [81, 84]. Here, we ought to mention that while sensing and estimating sleep metrics with consideration to bodily rhythms has been previously documented [9], we cannot assess with certainty the significance of menstrual and ovulation cycles in the computation of commercially available derived metrics due to the lack of peer-reviewed work and transparency in their documentation. Notably, in early 2025, OURA ring announced the introduction of menstrual cycle phases to the estimation of their metric of readiness [133]. Either way, there is a significant lack of documentation on how menstrual cycles relate to energy fluctuations, affecting both tracking practices and the interpretation of metrics.

Design Implication 2: To encourage cyclicity in the design of energy-relevant metrics and overcome this friction, we propose that design research on energy metrics should investigate and articulate recurring patterns to assist data contextualisation in daily variations.

Cyclicity is a common thread in all modes of energy tracking discussed in the previous section, whether for individuals managing energy-limiting conditions or for general health and wellbeing. Additionally, prior HCI research has also used crip theory to reason over cyclic pain and discomfort [99, 100] associated with menstrual cycles. Bianca [17] names this phenomenon in self-tracking "cyclic self-fashioning", framing how the datafication of menstrual cycles is used to make decisions within a day, for example, some cancel meetings during their premenstrual week. Tracking sports training in relation to seasonal changes has been also explored and proven promising in the ways that athletes expect their metrics to perform in relation to the external environment [144]. However, as our study has shown, felt energy of different individuals is affected by different cycles, and it is unclear how various body and energy cyclicities could be incorporated into the design of metrics. Prior research on felt experiences of bipolar condition has shown that co-design activities can be productively used to articulate and visualise personal bodily rhythms and make design choices that differ in form or format from those vetted by experts [121, 122]. As such, future research on energy tracking should acknowledge personal expertise using expressive modes of discussing cyclic temporality of energy.

7.2.3 Friction 3: Sensible Handling of "Unexpectedness" in Energy Tracking. We found many examples where our participants' bodies did not fit the models used by self-tracking technologies due to unexpected illness, stress, and physiological fluctuations in general. These incompatibilities led to frictions that reflect those experienced by people with ELCs [59, 114]. There is a notorious lack of diversity in datasets used to estimate body and health metrics [32, 46, 97], which may partly explain how normativities creep into health and energy estimations and recommendations, but it is not possible to ascertain the cause without access to the black box of metrics calculation. This friction goes beyond our prior recommendation for calibration and transparency, as well as the persistent argument in HCI to account for diversity [11, 12, 73]. Self-tracking and metric systems not only mis-interpret health states, they also reinforce temporalities of recovery from sickness that do not account for a diversity of bodies [143]. Assuming stability and uniformity even in cases of bodies being in flux or change is a phenomenon present in our findings and shared with other inquiries on tracking and aiding the body during fluctuations and transitions [24, 86]. The ideal of the stable body has also been challenged in HCI research on menstrual cycles that argue for embracing and celebrating the fluctuating body [21, 54, 102, 129]. Disability studies similarly argue that definitions of what is "normal" do not exist and therefore adapting bodies to universal types of medical and social models is rather paradoxical [108, 142].

The topic of irregularity and idiosyncratic health transitions was previously reported in relation to machine learning models in wellbeing, where authors suggested a shift towards the design for temporalities of recovery and restoration from illness [143]. In chronic condition management, a combination of a personalised model of health with contextual and crowdsourced data can be one potential resolution of responding to non-routine health fluxes in self-tracking [128].

Design Implication 3: All bodies fluctuate; thereby, design researchers should approach fluctuation as an expected phenomenon in the design of energy metrics.

There is more to be learned in self-tracking from abnormalities and unwellness rather than from "expected" normalcy. One way to do that is to move away from numerical types of metrics and rankings that relate to energy. Metrics associated with energy should follow Spiel's recommendation to move away from accuracy and instead consider *indicators* that invite openness in the interpretation of metrics [126]. Embracing the integration of qualitative and physicalised forms of data [58, 79] might resist the rigid categorisation of scores whilst individuals recover or transition to a new unexpected state. Similarly, ambiguous visualisations and actuations of metrics depict well the dynamic nature of bodies and, in the case of energy tracking, can introduce other somatic dimensions that are lost when thriving for accuracy [6, 112].

7.3 Turning to Dis/Ability with Crip Methodologies: What can Designers Learn About Energy-Tracking from Crip Theory?

Crip theory, originating in disability studies, informed the methodology used in this study. The concepts of "compulsory abledness" and dis/ability [134] inspired our approach to using the experiences of people with chronic conditions as a lens to study the self-tracking practices of people not currently experiencing chronic conditions. This is related to a key commitment of crip theory that acknowledges disabled individuals as "knowers and makers" and centres on expert crip knowledges as a valuable asset to society [93]. We attempted to resist uniformity in health definitions through our recruitment strategy and the design of the survey and interviews to surface how societal norms and biases shaped conceptions and uses of energy tracking and to welcome personal interpretations and meanings of energy, wellbeing, and care together with prior knowledges of illness and disability. We now discuss the potential

contentious or risky consequences of our approach of applying a crip approach to a non-disabled or chronically ill population, before highlighting the potential benefits of our approach that we propose outweigh these risks.

The aim of our research was to deploy crip approaches as a tactic to account for nuances in health and body definitions within a broader context of all bodies and abilities [90]. Crip theories were developed to address the subjugation of people with disabilities and chronic conditions [67]. As such, our extension of this theoretical framework to a group of people who do not currently experience chronic conditions or disabilities could be interpreted as an act of appropriation that might jeopardise its power to address the struggles of the originally intended group. The application of crip theory within a inquiry that does not actively study or benefit disabled and chronically ill individuals risks capitalising on, and eventually erasing, the efforts of underserved communities that are required to fight for their rights and against marginalisation. This is a similar argument held against the discipline of Universal Design that aims to address issues around accessibility through improving design for all [43]. Hamraie stresses that this approach risks erasing differences between bodies and abilities that leads to different groups of people having different needs [48]. They instead propose the shift of design practice to building accessibility collectively, identifying beneficial practices in communities and facilitating cross-pollination between these groups, without aiming for universality [49]. We also draw on this tactic of cross-pollination: between the expertise and experiences of people with chronic conditions, to those currently without. Unlike Universal Design, we do not recommend the design of one totalising energy tracking technology for all, both with and without chronic conditions. We support the design of dedicated pacing technologies specifically designed for people with energy limiting conditions. Rather, we offer implications for the design of normative fitness tracking technologies used outside of the context of chronic conditions that reflect valuable crip knowledges [93, 123].

Applying a crip approach facilitated the identification of societal norms embedded in energy conceptualisations and self-tracking practices outside of the practices of people with ELC and has subsequently led to design implications that acknowledge individual knowledges and consider fluctuations (expected and unexpected). We also demonstrate how the open-ness of crip methodologies fostered specifically knowledges of prior experiences of "irregularity" (e.g., illness, sickness) that trouble the dis/ability binary. We believe our design implications would improve the experience of all through supporting energy-tracking that reflects subjective, rather than optimisation-oriented, forms of energy-tracking. In light of increasing rates of workplace burnout and ELCs, this seems like a timely and important topic [4, 14]. By showing how societal norms and biases around health and disability affect everyone, our research aims to highlight how these same forces, in far more damaging ways, shape the lives of people with chronic conditions and disabilities [51]. The crip approach of this paper to energy tracking has often raised the question of who is interested in tracking their energy and, possibly, why, actively resisting the characterisation of energy-tracking as an optimising and technosolutionist practice that acknowledges energy fluctuations, just to optimise them to increase productivity [26, 104].

Altermark describes how vulnerability bridges the divide between "disability" and "normalcy" and is rooted in the fact that living our lives in bodies that are unpredictable and open to injury means that our ways of functioning can change [7]. Thereby, performing what they call a "crip solidarity" conveys that all bodies are entangled and reliant on institutions and social structures, and therefore, we all share responsibility to build and maintain. From this perspective, vulnerabilities, caused by the failure to fit societal and health standards or meet expectations, are not independent research problems but rather shared threads that travel across groups and identities. The goal of our research inquiry is not to take the spotlight away from a growing and important discourse on disability and the benefits of self-tracking, but to show that a crip orientation to self-tracking can illuminate how normativities get reproduced even for individuals who do not have any prevailing or ongoing disability, and, most importantly, that these normativities also re-enforce how people with disabilities are treated in society.

Interestingly, we are witnessing a rise of research at the intersection of underserved conditions and technology design [58, 83, 87, 141] that can serve as a promising ground for applications of the knowledge and implications of our study. With our conceptual scaffolding of energy tracking beyond chronic conditions, as shown in figures 2, 3, 4, our inquiry builds on a crip agenda that pushes the boundaries of disability studies to the point that perhaps even disability can become the rule, and normalcy the exception [7]. By doing so, we open a space for individuals who may not be marked as disabled, or are undergoing diagnosis, or live in liminal spaces of illness and wellness, but who nonetheless experience compulsory able-bodiedness. This is not a call to erase disabled knowledge or decentre disabled people; rather, it is a modest proposal to expand the boundaries of crip theory to include "unmarked" or unremarkable bodies —such as those undergoing diagnosis, fluctuating recovery, or simply accompanying the cyclical, ageing or often-times unpredictable fluctuations of the ever-changing bodies.

8 Reflexivity

All three authors use wearable self-tracking technologies, and two authors have had experiences managing energy and pacing. The first have been living with an enigmatic chronic condition for over 3 years, and second author have been managing their ELC for 5 years. Our experiences of living with chronic conditions assisted with the framing of the study and provided us with the tools to conduct a nuanced analysis of the invisible and enigmatic aspects of unwellness that might affect the lives of our participants [85, 94]. We also acknowledge the female over-representation in our interview materials, which we interpret, in light of the survey responses, as increased interest in energy tracking. As the findings present, menstrual and ovulatory phases impacted participants' energy levels and ability to recover, and therefore make energy an easier and more relevant phenomenon to notice and track. Therefore becoming a research topic participants' are more motivated to participate in. However, as we discussed, this is a hypothesis that requires further enquiry.

9 Future Work

This research study adds to the (incomplete) design space of energy tracking, opening the space beyond the scope of chronic conditions management. Future work should continue exploring alternative frameworks for understanding and tracking energy, especially in relation to the recent thematic map of interactions in energy tracking proposed by [116]. The conceptualisations of energy in our study challenge the finite view of energy often embedded in tracking metrics and instead encourages holistic approaches to the design of interactions with self-tracking devices that attune and care for the body [95, 96, 136], weaving together self-reporting methods with biodata sense-making practices, fostering self-knowledge rather than optimisation of health [60, 78, 111, 123]. Whereas our crip inquiry focused on qualitative content, energy as a material is another dimension that seems worth exploring. Designers might bring embodiment and the concept of energy as a material together [109] and explore entanglements with sensors, and trackers that represent alternative temporalities [108].

We invite future research to continue attending to the subject of rest through critical lenses [58, 63]. To mitigate surveillance and privacy issue that often rise with customisation and adaptability of systems [128], we once again turn to crip solidarity, proposing that future work utilises collective and open spaces as platforms to empower individuals through making [35, 50]. As our initial study was limited to a sample of users and their folk understandings of energy [105], future work should also delve into the computational and algorithmic dimensions of these metrics. By forming partnerships or by using open-source algorithms, we can better understand how these "black box" metrics are calculated and how they influence energy tracking behaviours.

10 Conclusion

The practice of tracking energy originates in chronic conditions management and pacing technologies. This paper explores energy conceptions and tracking practices outside the context of chronic conditions, but remains committed to dis/ability justice. Through a crip methodology, we show the relevance of energy tracking to individuals that are not currently managing chronic conditions and we identify three core energy conceptualisations informed by metrics: energy as a *reserve*, energy as a *measure of wellness* and energy as *cyclic* and ways in which energy tracking is used to manage and optimise energy levels and frictions arising between the designed and actual use of self-tracking technologies. We use Homewood's three modes of pacing to compare the practices of our participants to those of people with ELCs [55, 59] and we propose a conceptual scaffolding of energy tracking beyond the management of chronic conditions. We then contribute with three design recommendations to future research in energy tracking that address frictions arising in our inquiry. There are risks in the application of crip theories to groups that do not necessarily identify as disabled, though we demonstrate that crip theory can be a generative and rich lens to guide researchers' understandings of ability, bodily and health fluctuations, and normativities. We propose this extends its relevance and generative power beyond disability research, whilst still benefiting the lives of people with disabilities by raising awareness of their needs.

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